

Glossary of Acronyms

This guide uses many acronyms drawn from machine learning, systems engineering, and agent research. The following reference covers the most common ones; each entry appears in the chapter where the concept is first introduced in full.

A2A	Agent-to-Agent (communication protocol)	GEMM	General Matrix Multiplication
AdamW	Adam with decoupled Weight decay	GQA	Grouped Query Attention
BDI	Belief–Desire–Intention (agent architecture)	GRPO	Group Relative Policy Optimization
BERT	Bidirectional Encoder Representations from Transformers	HBM	High Bandwidth Memory
BLEU	Bilingual Evaluation Understudy (metric)	IPO	Identity Preference Optimization
BPE	Byte-Pair Encoding	KL	Kullback–Leibler (divergence)
BT	Bradley–Terry (preference model)	KTO	Kahneman–Tversky Optimization
CoT	Chain of Thought	KV	Key–Value (cache)
CTDE	Centralized Training, Decentralized Execution	LATS	Language Agent Tree Search
CUDA	Compute Unified Device Architecture	LLM	Large Language Model
DAG	Directed Acyclic Graph	LoRA	Low-Rank Adaptation
DAPO	Dynamic Adaptive Policy Optimization	LR	Learning Rate
DDP	Distributed Data Parallel	MCP	Model Context Protocol
DiLoCo	Distributed Low-Communication (training)	MCTS	Monte Carlo Tree Search
DP	Data Parallelism	MDP	Markov Decision Process
DPO	Direct Preference Optimization	MFU	Model FLOPS Utilization
DQN	Deep Q-Network	MHA	Multi-Head Attention
DRAM	Dynamic Random-Access Memory	MLA	Multi-head Latent Attention
ELO	Elo rating system (named after Arpad Elo)	MLP	Multi-Layer Perceptron
EOS	End of Sequence (token)	MoE	Mixture of Experts
FA	Flash Attention	NOOA	NVIDIA Object-Oriented Agents
FFN	Feed-Forward Network	NLL	Negative Log-Likelihood
FLOP	Floating-Point Operation	OPSD	On-Policy Self-Distillation
FSDP	Fully Sharded Data Parallel	ORPO	Odds Ratio Preference Optimization
GAE	Generalized Advantage Estimation	ORM	Outcome Reward Model
GAIA	General AI Assistants (benchmark)	PEFT	Parameter-Efficient Fine-Tuning
		PP	Pipeline Parallelism
		PPO	Proximal Policy Optimization
		PRM	Process Reward Model
		QLoRA	Quantized Low-Rank Adaptation
		RAG	Retrieval-Augmented Generation
		ReAct	Reasoning + Acting (agent pattern)
		RL	Reinforcement Learning

RLHF	Reinforcement Learning from Human Feedback	SM	Streaming Multiprocessor
RLVR	Reinforcement Learning with Verifiable Rewards	SPLADE	SParse Lexical AnD Expansion
RM	Reward Model	SRAM	Static Random-Access Memory
RoPE	Rotary Position Embedding	SSE	Server-Sent Events
ROUGE	Recall-Oriented Understudy for Gisting Evaluation	SWE-bench	Software Engineering Benchmark
RRF	Reciprocal Rank Fusion	TD	Temporal Difference (learning)
SDK	Software Development Kit	TP	Tensor Parallelism
SFT	Supervised Fine-Tuning	TRL	Transformer Reinforcement Learning (library)
SGD	Stochastic Gradient Descent	UCB	Upper Confidence Bound
		vLLM	Virtual LLM (inference engine)